

Looking for a Tunnell-style formula

The square root of $L(E_p, 1)/\Omega$ for $y^2 = x^3 + px$, what governs its parity, and what does not
sequel to III of $y^2 = x^3 + px$ at the Fermat primes

1. What a Tunnell-style formula is

The companion note reduced the Fermat-prime pattern to a single assertion,

$$L(E_p, 1)/\Omega = 2m^2 \quad \text{with} \quad m = 2^{k-2},$$

and observed that $L(1)/\Omega$ is **twice a perfect square** in all 64 computed cases. A square is the signature of a particular kind of theorem, and this section says which.

1.1. Half-integral weight and the Shimura correspondence

Classical modular forms have integral weight. There is a parallel world of forms of **half-integral** weight $k + 1/2$, and Shimura constructed a correspondence

$$\text{weight } k + 1/2 \quad \longleftrightarrow \quad \text{weight } 2k$$

matching Hecke eigenforms on the two sides. The half-integral side is smaller and stranger — its Fourier coefficients are indexed by integers but behave like **square roots** of the quantities the integral side sees. Concretely, if $g = \sum c(n)q^n$ has weight $3/2$ and corresponds to a weight-2 form f , then $c(n)$ knows about f twisted by the quadratic character of n .

1.2. Waldspurger's theorem

Waldspurger made that precise, and in the form we care about it reads: for f of weight 2 with a weight- $3/2$ partner $g = \sum c(n)q^n$,

$$L(f \text{ twisted by } \chi_d, 1) = \kappa \cdot \frac{c(|d|)^2}{\sqrt{|d|}}$$

for d ranging over a fixed square class, with κ independent of d . **The central value of the twisted L -function is the square of a Fourier coefficient.** That is where squares come from, and it is why such formulas are so effective: a coefficient $c(|d|)$ is a finite, computable integer, so vanishing of a central L -value — an analytic condition — becomes the arithmetic condition $c(|d|) = 0$.

1.3. Tunnell's theorem, the model case

The famous instance is the congruent number problem, $E_n : y^2 = x^3 - n^2x$. Tunnell (1983) identified the weight- $3/2$ partner explicitly, and its coefficients count representations by ternary quadratic forms. For n odd and squarefree, put

$$a_n = \#\{(x, y, z) \in \mathbb{Z}^3 : n = 2x^2 + y^2 + 32z^2\} - 2\#\{(x, y, z) \in \mathbb{Z}^3 : n = 2x^2 + y^2 + 8z^2\}.$$

Then

$$L(E_n, 1) = \frac{\Omega a_n^2}{4\sqrt{n}},$$

so that n congruent $\implies a_n = 0$, and conversely under BSD. The whole analytic content is compressed into counting lattice points on two ellipsoids.

Why this matters for III. Under BSD the order of III is $L(1)/\Omega$ times torsion and Tamagawa factors. If $L(1)/\Omega$ is essentially $c(|d|)^2$, then $\#\text{III}$ is essentially a square — which is exactly what the Cassels–

Tate pairing independently demands. The two statements meet, and m is the arithmetic square root that both are pointing at.

2. Why our family is not literally Tunnell's

Here is the obstacle. Waldspurger's theorem governs **quadratic** twist families. Ours is not one.

The quadratic twist of $y^2 = x^3 + dx$ by t is $y^2 = x^3 + dt^2x$, so twisting moves d inside its **square** class. Two squarefree integers $d \neq d'$ are therefore never quadratic twists of each other: the family $\{E_d\}_{d \text{ squarefree}}$ is the family of **quartic** twists of $y^2 = x^3 + x$, indexed by $\mathbb{Q}^\times/(\mathbb{Q}^\times)^4$, and each quadratic twist class contains just one squarefree d .

What does organise the family is the complex multiplication. E_d has CM by $\mathbb{Z}[i]$, and

$$L(E_d, s) = L(s, \bar{\psi}\chi_d),$$

a Hecke L -function of $\mathbb{Q}(i)$, where $\chi_d = (d/\cdot)_4$ is the **quartic** residue character. At a prime $p = \pi\bar{\pi}$ of good reduction,

$$a_p = 2 \operatorname{Re} \left(\overline{\chi_d(\pi)}\pi \right).$$

So the family is a family of quartic twists of one Hecke character. The square shape of the central value still has an explanation on this side — values of such L -functions at the centre are, via elliptic units, essentially norms from $\mathbb{Q}(i)$, and a norm is a square in the relevant sense — but it is not Waldspurger's theorem, and the coefficient m is not a Tunnell coefficient.

Nevertheless the computation says a square root exists and is small: $L(1)/\Omega = 2m^2$ in every one of 64 cases, with $m = 1, 2, 3, 4$ occurring. So there **is** something to find. The rest of this note is what we could determine about m .

3. The first rung: when is m even?

This one has a clean answer, and it is classical arithmetic rather than modular forms.

Observation 1. For every rank-0 prime $p \equiv 1 \pmod{8}$ tested (82 of them, $p < 4000$):

$$m \text{ is even} \iff 2 \text{ is a quartic residue mod } p \iff 8 \mid b,$$

where $p = a^2 + b^2$ with a odd. Zero exceptions.

The second equivalence is Gauss's criterion, and the third description is the classical

$$2 \text{ a quartic residue mod } p \iff p = x^2 + 64y^2.$$

For a Fermat prime this settles the bottom of the pattern immediately. Since $p = 2^{2^k} + 1 = 1^2 + (2^{2^{k-1}})^2$ we have $a = 1$ and $b = 2^{2^{k-1}}$, so

$$8 \mid b \iff 2^{k-1} \geq 3 \iff k \geq 3.$$

And indeed $m = 1$ at $k = 2$, $m = 2$ at $k = 3$, $m = 4$ at $k = 4$: odd exactly when $k = 2$.

4. The second rung is not the octic criterion

The obvious continuation — $4 \mid m$ iff 2 is an **octic** residue mod p — is false, and badly so.

Observation 2. Among the 101 primes $p < 20000$ with m even, the condition $4 \mid m$ disagrees with “2 is an octic residue mod p ” in **48** cases.

A clean counterexample pair, which also kills the guess that $p = x^2 + 256y^2$ is the criterion:

p	a	b	$p = x^2 + 256y^2$?	m
4177	9	64	$9^2 + 256 \cdot 4^2$	2
4937	29	64	$29^2 + 256 \cdot 4^2$	4

Same b , same $v_2(b) = 6$, both represented by $x^2 + 256y^2$ with the same y — and m differs. Whatever governs the higher 2-adic valuation of m is not a power-residue condition on 2 alone.

5. Status, and where to look next

Established here. The background reduction: a Tunnell-style formula is exactly a statement that the central value is a square of a half-integral-weight coefficient, and that is the shape our $L(1)/\Omega = 2m^2$ has. The obstruction to importing Waldspurger directly: our family is a **quartic** twist family, so the right framework is the Hecke character $\overline{\psi}\chi_d$ with χ_d quartic, not a weight-3/2 form. And Observation 1, which pins the parity of m to Gauss’s quartic residue criterion and thereby explains the bottom of the Fermat pattern.

Refuted here. The natural continuation by octic residues (Observation 2).

Still open. The full statement $m = 2^{k-2}$. The place we would look next is the literature on the **2-adic valuation of central L -values** in exactly these CM families — Zhao’s criteria for $y^2 = x^3 - dx$ and Tian’s induction on the number of prime divisors for the congruent number problem. Those compute $v_2(L(1)/\Omega)$ from combinatorial data attached to the prime factors of d (residue symbol graphs, and the rank of an associated matrix over $\mathbb{F}_2$). For $d = p$ prime that data degenerates to a single vertex, which is why one might hope the Fermat case is within reach — and why it is strange that it grows at all.

The oddity worth stating plainly. For the Fermat primes, $m = 2^{k-2}$ while $b = 2^{2^{k-1}}$. So m is **doubly logarithmic** in b :

$$\log_2 m = (\log_2 \log_2 b) - 1.$$

No algebraic expression in a and b behaves like that. Whatever computes m must be a counting or class-number-like quantity, of the sort that is generically of size $\sqrt{p}$ but collapses to something tiny on a thin family — which is precisely the character of a half-integral weight coefficient, and is the best evidence we have that a Tunnell-style formula is the right thing to be looking for.